General Astronomy

1. $D = 10\,m$
$\lambda = 550\,nm$
$= 550 \times 10^{-9}\,m$

$\therefore$ θ/Angular res

$= 1.22 \frac{\lambda}{D}$

$= 1.22 \times \frac{550 \times 10^{-9}}{10}$

$= 1.22 \times 550 \times 10^{-10}$

$= 12.2 \times 55 \times 10^{-10}$

$= 122 \times 55 \times 10^{-11}$

$= 6710 \times 10^{-11}$

$= 671 \times 10^{-10}$

$= \boxed{0.671 \times 10^{-7}}$

Q2 Radio Telescopes V/s Optical Telescopes

Both radio and optical telescopes are essential tools in the field of astronomy but have varied usages in the different ranges of the spectrum.

Comparison Table

Feature	Radio Telescopes	Optical Telescopes
Wavelength detected	Radio waves (long wavelength)	visible Light
Weather Sensitivity	Can operate in cloudy, rainy or dust cond^n s	Limited by clouds, weather, and atmospheric turbulence
Day/Night operation	Can be used day and Night	mostly limited to night time us
Interference	Prone to radio freq. Interference from electronics	Prone to light poll. from artificial sources
Resolution	Lower (requires a lot of setup)	Higher (for a given aperture size)
Construction Size	Typically very large to collect weak signals	Can be smaller for similar resolution
Penetration of dust	Can see through cosmic dust of clouds	Blocked by dust, cannot see through dense regions
Image Production	Requires complex data processing	Direct visual images possible
Cost	Expensive due to size and Technology	varies, large professional scopes are costly.

Advantages

Radio

- All weather, all-time operation
- Penetrate dust clouds
- Study unique phenomena
- Interferometry

Optical

- High resolution
- Direct Imaging
- Rich historical data
- Versatility.

Limitations

Radio

- Low Resolution
- Radio Interference
- Complex data Processing
- High cost and size

Optical

- Weather and Atmosphere
- Limited to night
- Obscured by dust
- Light Pollution

Example Use-Cases

Radio

- Mapping H_2 gas in galaxies
- Studying pulsars and quasars
- Observing cosmic microwave
- Imaging through dust clouds

Optical

- Imaging planets, stars and galaxies
- Discovering Exoplanets
- Amateur astronomy
- Space based telescopes (e.g. Hubble).

(Q3) The Event horizon is the boundary surrounding a black hole. It marks the limit where the gravitational pull becomes so strong that nothing-not even light-can escape. This is why black holes appear "black": ~~any matter~~ any matter or radiation that crosses this boundary is lost to the outside universe forever.

Some Characteristics:-

- Not a physical surface
- Escape velocity:

what happens to the space-time ~~warping~~ at the Event horizon

- Extreme warping: The presence of a black hole curves space-time to an extreme degree. At the event horizon, this curvature becomes so intense that all possible paths an object could take lead inward, towards the black hole center (the singularity).

- Time dilation: As an object approaches the event horizon, time appears to slow down for it from the perspective of a distant observer.

- Causality:- Once inside the event horizon, all future paths are directed towards the singularity. No info/signal can back out, fundamentally altering the causal structure of space-time

Point of No-Return

- Irreversible crossing: Once crossed, escaping is impossible
- Hidden Events: Any event that occurs inside the event horizon cannot influence the outside universe

- Singularity destination:- all matter and energy that cross the event horizon are drawn to it

CSE / AIML or its Allied Branches

(Q1)

Retrograde motion in astronomy refers to the apparent backward movement of a planet or celestial body across the sky, relative to the background stars. This phenomenon is most commonly observed with planets in our Solar System and is an optical illusion caused by the relative positions and motions of Earth and other planet as they orbit the Sun